

for individual communication with instructor and fellow students and for submitting assignments. Assignments may be submitted as text-based e-mail, voice-based mail, Word or Text. Although one can use the various ready-made software packages available in the market for the purpose, the prototype of this framework is developed to demonstrate the powerful features of Web services-based architecture. The prototype Web service code is written in Java programming language. All the communication is done through midlets.

Developing this Web service model is unique from other software development. The end user is usually known prior to development as well as the device/application that will be used. In this case, there was no specifically defined user. Therefore multiple devices and consumers had to be considered. From a development standpoint, the challenge was to build a dictionary database.

There could be a number of different ways in which the m-learning model could be implemented:

1. Mobile Internet
 2. Online access via mobile telephone to the entire course
 3. 'Download-on-demand' version
- We have used the mobile Internet

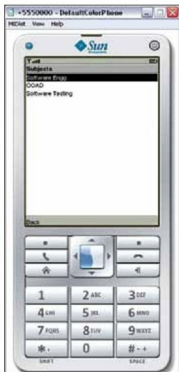


Fig. 4: Subject list

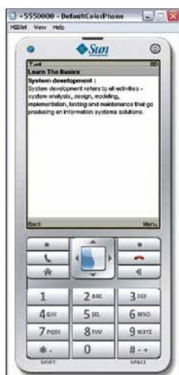


Fig. 5: Learning

service model.

To realise a platform-independent application which can be used on a variety of different operating systems for smart phones, a standardised development environment, for example, the Java 2 Micro Edition (J2ME), is necessary. Almost every mobile phone to date is Java-enabled. Subsequently, these are capable of executing a J2ME application. J2ME is platform-independent and makes the creation of mobile, Web-based applications possible. The sample screenshots are shown in Figs 2 through 5.

Validation and results

The various aspects of the prototype considered for validation were:

1. **User interface transparency:** This was tested by having multiple models of mobile phones and checking whether the contents are displayed uniformly without any change in the software code.
2. **Security:** Authentication and encryption prevent unauthorised users from accessing the learning content even if they possess the device physically.
3. **Application transparency:** The applications hide the underlying hardware from the users and provide a

continuous environment for access and management. The nitty-gritty of the hardware and network are completely abstracted from the user.

We achieved a reasonable degree of success in the above properties. The tools presented can be used within any field of work, be it English, mathematics or any other content. Therefore students can use the same platform and application for studying several subjects.

Future possibilities

While mobile phones can provide the necessary support for the organisational changes of learning, it remains to be seen whether the pedagogical benefits are sufficiently convincing. However, the phenomenal growth of mobile computing will increase the need for future research, especially in the area of human-computer interaction and media psychology with focus on adaptive and sensitive interfaces and adapted content.

Research can be taken up to explore ways to create mobile content on the fly and add intelligence to the system. Voice interaction with the mobile device will allow students and teachers to create annotations only by speech. The use of different media, such as video or audio files, is already being studied. Integration of different applications such as Web browsers could allow users to fetch additional information directly from the Web and include that in the learning content and applications.

In addition, more experiments could be done to examine the effectiveness and potential of mobile devices for direct use in the teaching and learning process and not only as a supporting tool. Studies of this kind are still rare, since this field is relatively new. It provides the foundation for further development and expansion of mobile devices integration in higher educational systems. ●

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